

Geometrical Representation:- To draw curve of complex variable (x, iy) we take two axes i.e., one real axis and the other imaginary axis. A number of points (x, iy) are plotted on z -plane, by taking different values of z (different values of x and y). The curve C is drawn by joining the plotted points. The diagram obtained is called Argand diagram.

$$\text{Let } w = f(z) = f(x + iy) = u + iv$$

To draw a curve of w , we take u -axis and v -axis. By plotting (u, v) on w plane and joining them, we get a curve C_1 on w plane.

Transformation:- For every point (x, y) in the z plane, the relation $w = f(z)$ defines a corresponding point (u, v) in the w -plane. We call this transformation or mapping of z plane into w plane. If a point z_0 maps into the point w_0 it is also known as the image of z_0 .

If the point $P(x, y)$ moves along a curve C in z -plane, the point $P(u, v)$ will move along a corresponding curve C_1 in w -plane. We then say that a curve C in the z plane is mapped into the corresponding curve C_1 in the w plane by the relation $w = f(z)$.

Ex: Transform the curve $x^2 - y^2 = 4$ under the mapping $w = z^2$

$$\text{sol:- } w = z^2 = (x + iy)^2 \Rightarrow u + iv = x^2 - y^2 + 2ixy$$

$$\text{Thus gives } u = x^2 - y^2 \text{ and } v = 2xy$$

Conformal transformation:- Let two curves C, C_1 in the z -plane intersect at the point P and the corresponding curve C', C'_1 in the w -plane intersect at P' . If the angle of intersection of the curves at P in z -plane is the same as the angle of intersection of the curve of w -plane at P' in magnitude and sense, then the transformation is called conformal.

Conditions: (i) $f(z)$ is analytic (ii) $f'(z) \neq 0$ or If the sense of rotation as well as the magnitude of ~~the~~ the angle is preserved the transformation is said to be conformal.

If ~~the sense of~~ only the magnitude of the angle is preserved transformation is Isogonal

Note:- If $f(z)$ is analytic and $f'(z) \neq 0$ in a region R of the z -plane then the mapping $w = f(z)$ is conformal at all points of R

Some standard transformations:-

(1) Translation:- $w = z + C$, where C is a complex constant

Let $z = x + iy$, $C = a + ib$ and $w = u + iv$ then the transformation

becomes $u + iv = (x + iy) + (a + ib) = (x + a) + i(y + b)$

So that $u = x + a$, $v = y + b$

Thus the transformation is a mere translation of the axes and preserves the shape and size

(2) Rotation and Magnification:- $w = CZ$, where C is complex constant

Let $C = p e^{i\alpha}$, $z = r e^{i\theta}$ and $w = R e^{i\phi}$ then the transformation

becomes $R e^{i\phi} = p r e^{i(\theta + \alpha)}$

So that $R = pr$ and $\phi = \theta + \alpha$

Thus the transformation maps a point (r, θ) in the z -plane into a point $P'(pr, \theta + \alpha)$ in the w -plane.

(3) Inversion: $w = \frac{1}{z}$ Let $z = r e^{i\theta}$ and $w = R e^{i\phi}$ then the transformation

becomes $R e^{i\phi} = \frac{1}{r} e^{-i\theta}$ so that $R = \frac{1}{r}$ and $\phi = -\theta$

Thus under the transformation ~~becomes $R e^{i\phi} = \frac{1}{r} e^{-i\theta}$~~ $w = \frac{1}{z}$, a point $P(r, \theta)$ in z -plane is mapped into the point $P(\frac{1}{r}, -\theta)$

4. Bilinear transformation:- A transformation of the form

$w = \frac{az + b}{cz + d}$ where a, b, c, d are complex constants and

$ad - bc \neq 0$ is a bilinear or Mobius transformation

The transformation given by (1) is conformal, since

$\frac{dw}{dz} = \frac{ad - bc}{(cz + d)^2} \neq 0$, The inverse mapping of (1) is $z = \frac{-dw + b}{cw - a}$

which is also a bilinear transformation

Note 1: The cross-ratio is invariant under a bilinear transformation. Thus w_1, w_2, w_3 and w_4 are the respective images of four distinct points z_1, z_2, z_3 and z_4 then

$$\frac{(w_1 - w_3)(w_2 - w_4)}{(w_1 - w_4)(w_2 - w_3)} = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_4)(z_2 - z_3)}$$

Note 2: In the bilinear transformation $w = \frac{az+b}{cz+d}$, $ad-bc \neq 0$ dividing the numerator and denominator of the right hand side by one of the four constants, we observe that there are only three independent constants. Hence three independent conditions are required to determine a bilinear transformation.

1. Consider the transformation $w = e^{i\frac{\pi}{4}} z$ and determine the region in the w -plane corresponding to the triangular region bounded by the lines $x=0$, $y=0$ and $x+y=1$ in the z -plane.

Sol The given transformation is $w = u+iv = \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)(x+iy)$

$$= \frac{1}{\sqrt{2}}(1+i)(x+iy)$$

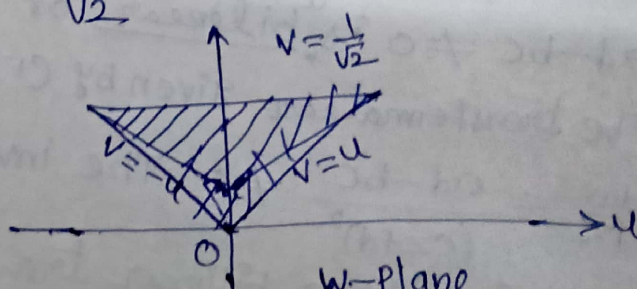
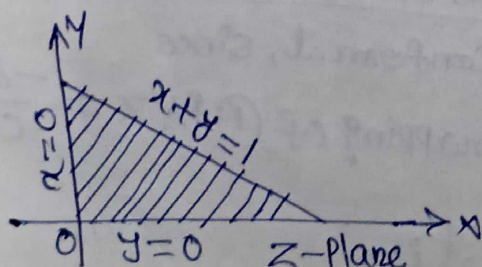
$$= \frac{1}{\sqrt{2}}[(x-y) + i(x+y)]$$

$$u = \frac{1}{\sqrt{2}}(x-y) \text{ and } v = \frac{1}{\sqrt{2}}(x+y)$$

The lines $x=0$ maps into $u = -\frac{1}{\sqrt{2}}y$, $v = \frac{1}{\sqrt{2}}y$ or into $v = -u$

The line $y=0$ maps into $u = \frac{1}{\sqrt{2}}x$, $v = \frac{1}{\sqrt{2}}x$ or into $v = u$

The line $x+y=1$ maps into $v = \frac{1}{\sqrt{2}}$



2 Find the image of the infinite strip $\frac{1}{4} \leq y \leq \frac{1}{2}$ under the transformation $w = \frac{1}{z}$. Also show that the regions graphically.

Sol The given transformation is

$$w = \frac{1}{z} \text{ or } z = \frac{1}{w} \text{ or } x + iy = \frac{1}{u + iv} = \frac{u - iv}{u^2 + v^2}$$

So that $x = \frac{u}{u^2 + v^2}$ and $y = \frac{-v}{u^2 + v^2}$

If $y = \frac{1}{4}$ then $\frac{-v}{u^2 + v^2} = \frac{1}{4}$

$$\Rightarrow u^2 + v^2 + 4v = 0 \text{ or } u^2 + (v+2)^2 = 4$$

or

If $y = \frac{1}{2}$, then $\frac{-v}{u^2 + v^2} = \frac{1}{2}$ or $u^2 + v^2 + 2v = 0$ or $u^2 + (v+1)^2 = 1$

Hence the infinite strip $\frac{1}{4} \leq y \leq \frac{1}{2}$ is transformed into the region between the two circles

$$u^2 + (v+2)^2 = 4, \text{ centre } (0, -2), \text{ radius } 2$$

$$\text{and } u^2 + (v+1)^2 = 1, \text{ centre } (0, -1), \text{ radius } 1$$

3 Discuss the transformation $w = \sin z$

Sol The given transformation is $w = u + iv = \sin(x + iy)$

$$\text{or } u + iv = \sin x \cosh y + i \cos x \sinh y$$

So that $u = \sin x \cosh y$ and $v = \cos x \sinh y$ — (1)

Eliminating y from equation (1), we get

$$\frac{u^2}{\sin^2 x} - \frac{v^2}{\cos^2 x} = 1 \quad (\because \cosh^2 y - \sinh^2 y = 1)$$

Thus the straight lines $x = c$ in the z -plane are mapped into confocal hyperbolas in the w -plane.

Eliminating x from equation (1), we get $\frac{u^2}{\cosh^2 y} + \frac{v^2}{\sinh^2 y} = 1$

Thus the straight lines $y = c$ in the z -plane are mapped into confocal ellipses.

The lines $x = 0$, and $y = 0$ map into the lines $u = 0$ and $v = 0$ respectively in the w -plane.

4 Find the image of the circle $|z-1|=1$ in the complex plane under the mapping $w = \frac{1}{z}$

sol $w = \frac{1}{z}$ or $u+iv = \frac{1}{x+iy} = \frac{x-iy}{x^2+y^2}$

$$u = \frac{x}{x^2+y^2}, v = \frac{-y}{x^2+y^2}$$

Given $|z-1|=1$ or $|x+iy-1|=1$ or $(x-1)^2+y^2=1$

$$x^2+y^2-2x=0 \text{ or } x^2+y^2=2x \text{ or } \frac{1}{2} = \frac{x}{x^2+y^2} = u$$

$$\frac{1}{2} = u \text{ or } 2u-1=0$$

5 Find the bilinear transformation which maps the points $z=1, i, -1$ into the points $w=i, 0, -i$. Hence find the image $|z|<1$

sol Let the required bilinear transformation be $w = \frac{az+b}{cz+d}$ — (1)

Substituting the corresponding values of w and z in (1), we get

$$i = \frac{a+b}{c+d}, 0 = \frac{a+ib}{c+id}, -i = \frac{-a+b}{-c+d}$$

These equations can be written as

$$(a+b) - i(c+d) = 0 \text{ — (2)} \quad b+ia=0 \text{ — (3)} \quad (-a+b) + i(-c+d) = 0 \text{ — (4)}$$

From (3) $b = -ia$

Adding (2) and (4), $2b - 2ic = 0$ or $c = \frac{b}{i} = -a$

Subtracting (4) from (2), $2a - 2id = 0$ or $d = \frac{a}{i} = -ia$

Substituting for b, c, d in (1), we get $w = \frac{az-ia}{-az-ia}$ or $w = \frac{1-z}{i+z}$ (5)

which is the required bilinear transformation

Now from (5) $z = i \frac{1-w}{1+w}$

$\therefore |z|<1$ is mapped into the region

$$\left| i \frac{1-w}{1+w} \right| < 1 \text{ or } \frac{|i||1-w|}{|1+w|} < 1 \text{ or } |1-w| < |1+w|$$

$$\text{or } |1-u-iv| < |1+u+iv| \text{ or } (1-u)^2+v^2 < (1+u)^2+v^2$$

Hence the interior of the circle $x^2+y^2=1$ in the z -plane is mapped into the entire half of the w -plane to the right of the imaginary axis

P Find the bilinear transformation that maps the points $0, 1, i$ in z -Plane onto the points $1+i, -i, 2-i$ in the w -Plane.

sol The required bilinear transformation is

$$\frac{(w_1 - w_2)(w_3 - w)}{(w_1 - w)(w_3 - w_2)} = \frac{(z_1 - z_2)(z_3 - z)}{(z_1 - z)(z_3 - z_2)}$$

$$\frac{(1+i-i)(2-i-w)}{(1+i-w)(2-i+i)} = \frac{(0-1)(i-z)}{(0-z)(i-1)}$$

$$\Rightarrow \frac{(1+2i)(2-i-w)}{2(1+i-w)} = (i-1) \left(\frac{i-z}{z} \right)$$

$$\frac{2-i-w}{1+i-w} = \frac{2(3i+1)}{5} \left(\frac{i-z}{z} \right)$$

solving for w ,

$$5z(2-i-w) = 2(3i+1)(1+i-w)(i-z)$$

$$\text{or } w = \frac{(6i+2)(1+i)(i-z) - (2-i)5z}{-5z + (6i+2)(i-z)}$$

$$w = \frac{z(6+3i) + (8+4i)}{z(7+6i) + (6-2i)}$$

P Determine the linear fractional transformation that sends the points $z=0, -i, 2i$ into the points $w=5i, \infty, -\frac{i}{3}$ respectively.

$z_1 \quad z_2 \quad z_3$

$w_1 \quad w_2 \quad w_3$

$$\frac{(0+i)(2i-z)}{(0-z)(2i+i)} = \frac{(5i-w_2)(-\frac{i}{3}-w)}{(5i-w)(-\frac{i}{3}-w_2)}, \text{ where } w_2 = \infty$$

$$\text{so } \frac{z-2i}{3z} = \lim_{w_2 \rightarrow \infty} \frac{(\frac{5i}{w_2}-1)(\frac{i}{3}+w)}{(5i-w)(-\frac{i}{3w_2}+1)}$$

$$\frac{z-2i}{3z} = \frac{\frac{i}{3}+w}{w-5i}$$

$$\frac{z}{3z} - \frac{2i}{3z} = \frac{\frac{i}{3} + w}{w - 5i}$$

$$\frac{1}{3} - \frac{2i}{3z} = \frac{\frac{i}{3} + w}{w - 5i}$$

$$\Rightarrow -\frac{2i}{3z} = \frac{\frac{i}{3} + w}{w - 5i} - \frac{1}{3}$$

$$\Rightarrow \frac{2i}{z} = \frac{-i - 3w}{w - 5i} + 1$$

$$\frac{2i}{z} = \frac{-i - 3w + w - 5i}{w - 5i} = \frac{-2w - 6i}{w - 5i}$$

$$\frac{2i}{z} = \frac{2w + 6i}{5i - w}$$

$$\text{or } (5i - w)2i = (2w + 6i)z$$

$$-10 - 2wi = 2wz + 6iz$$

$$-2wi - 2wz = 6iz + 10$$

$$-2w(i + z) = 6iz + 10$$

$$\Rightarrow w = \frac{6iz + 10}{-2(i + z)} = \frac{-3iz - 5}{z + i} = \frac{-3z + 5i}{-iz + 1}$$

b what are the invariant points of this transformation. Find the image of $|z| < 1$

$$\text{so } w = z = \frac{-3iz - 5}{z + i} \Rightarrow \frac{z}{z + i} + zi + 3zi + 5 = 0$$

$$\Rightarrow \frac{z}{z + i} + 4zi + 5 = 0$$

~~zz~~

$$z = -\frac{4i \pm 6i}{2} = i, -5i \text{ are the invariant points}$$

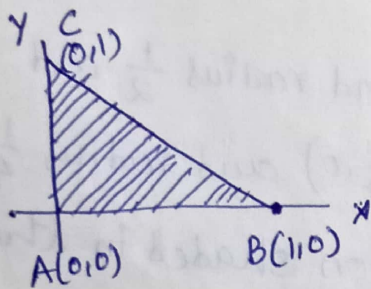
Fixed or invariant points:

Fixed points of function $w = f(z)$ are points which are mapped onto themselves i.e., $w = f(z) = z$

P Find the plot the image of triangular region with vertices at $(0,0)$, $(1,0)$, $(0,1)$ under the transformation $w = (1-i)z + 3$

sol $u + iv = w = (1-i)(x + iy) + 3$

so $u(x,y) = x + y + 3$, $v(x,y) = y - x$



AB: $y = 0$, $u = x + 3$, $v = -x$

or $u = -v + 3$

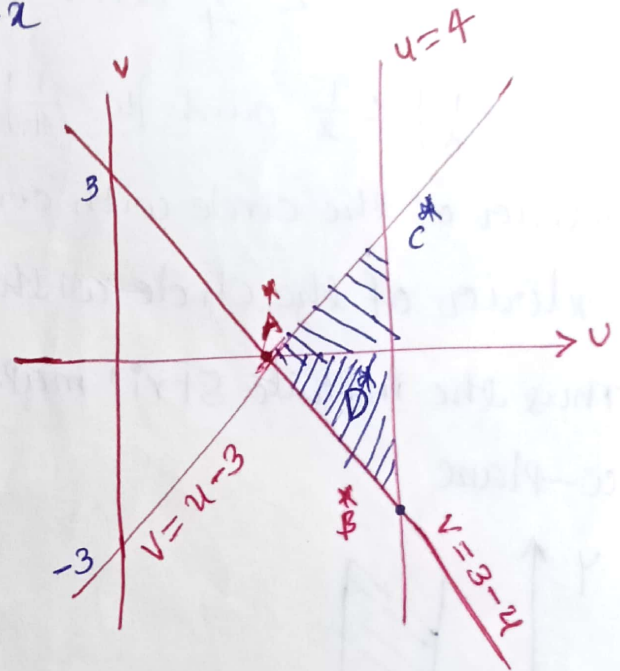
$\therefore v = 3 - u$: $A^* B^*$

AC: $x = 0$, $u = y + 3$, $v = y$

or $u = v + 3$ $\therefore v = u - 3$: $A^* C^*$

BC: $x + y = 1$ or $u = x + y + 3$

$u = 4$ $B^* C^*$



So, the image is the triangular region with vertices at $A^*(3,0)$, $B^*(4,-1)$, $C^*(4,1)$. Let $D = (\frac{1}{4}, \frac{1}{4})$ be any interior point of ABC. Its image is $D^*(3.5, 0)$ which is also interior of $A^* B^* C^*$

P Find the graph the strip $1 < z < 2$ under the mapping $w = \frac{1}{z}$

sol $u + iv = w = \frac{1}{z} = \frac{x - iy}{x^2 + y^2}$

$$\text{So } u = \frac{u}{u^2+v^2}, \quad v = \frac{-v}{u^2+v^2}$$

$$\text{Since } 1 < u < 2 \quad \text{so } 1 < \frac{u}{u^2+v^2} < 2$$

$$\text{or } u^2+v^2-u < 0 \quad \text{and} \quad 2(u^2+v^2)-u > 0$$

Rewriting

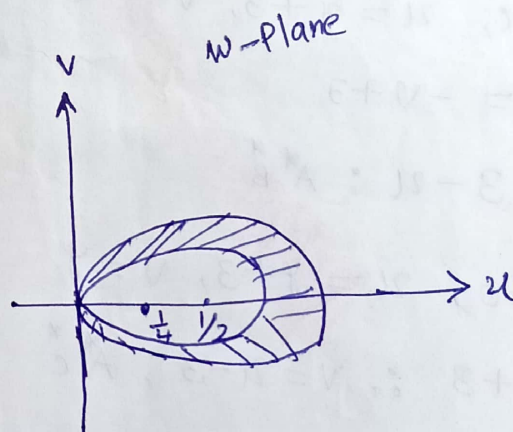
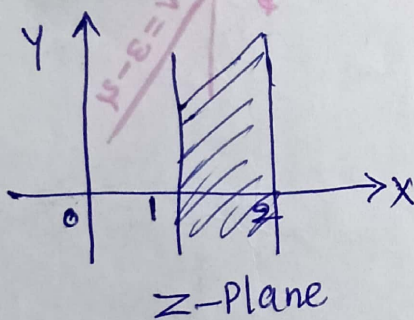
$$(u-\frac{1}{2})^2+v^2 < \frac{1}{4} \quad \text{and} \quad (u-\frac{1}{4})^2+v^2 > \frac{1}{16}$$

$$\text{or } |w-\frac{1}{2}| < \frac{1}{2} \quad \text{and} \quad |w-\frac{1}{4}| > \frac{1}{4}$$

interior of the circle with centre at $(\frac{1}{2}, 0)$ and radius $\frac{1}{2}$ and

exterior of the circle with centre at $(\frac{1}{4}, 0)$ and radius $\frac{1}{4}$

Thus the infinite strip maps to the region shaded in the w -plane



- Q Define conformal mapping, Find all points at which the mapping $w = z^2 + \frac{1}{z^2}$ are not conformal
- Q Find the linear fractional transform that maps the point $(-2, 0, 2)$ on to the points $(\infty, \frac{1}{2}, \frac{3}{4})$
- Q Determine a and b such that the function $f(z) = az^3 + bz^2$ is harmonic

p $z = 1, i, -1$ onto $w = 2, i, -2$
 $z_1 \quad z_2 \quad z_3 \quad w_1 \quad w_2 \quad w_3$

$$\frac{(w_1 - w_2)(w_3 - w)}{(w_1 - w)(w_3 - w_2)} = \frac{(z_1 - z_2)(z_3 - z)}{(z_1 - z)(z_3 - z_2)}$$

$$\frac{(2-i)(-2-w)}{(2-w)(-2-i)} = \frac{(1-i)(-1-z)}{(1-z)(-1-i)}$$

$$\Rightarrow \frac{(2-i)(-2-w)}{(2-w)(-2-i)} = \frac{-4-2w+2i+wi}{-4-2i+2w+wi} \cdot \frac{-1-z+i+zi}{-1-i+z+zi}$$

$$+4+4i-4z-4zi+2w+2wi-2wz-2wzi-2i-2i^2+2iz$$

$$+2zi^2-wi-wi^2+wi^2+wi^2+wi^2$$

$$= 4+4z-4i-4zi+2i+2iz-2i^2-2i^2z-2w-2wz+2wi+2wz$$

$$-wi-wzi+wi^2+wi^2$$

$$\Rightarrow 4i-4z+2w-2wz-2wzi-2i+2zi^2-wi^2+wi^2z$$

$$= 4z-4i+2i-2i^2z-2w-2wz+wi^2+wi^2z$$

$$\Rightarrow 2i-6z+3w-2wz+wi^2z = 6z-2i-3w-2wz+wi^2z$$

$$\Rightarrow 4i-12z+6w+2wi^2z=0$$

$$\Rightarrow 4i-12z+2w(3+i^2z)=0$$

$$\Rightarrow 2w(3+i^2z) = 12z-4i \Rightarrow w = \frac{12z-4i}{2(3+i^2z)} = \frac{6z-2i}{3-i^2z}$$

$$\therefore w = \frac{6z-2i}{i^2z-3}$$

p $z = \infty, i, 0$ onto $w = 0, i, \infty$
 $z_1 \quad z_2 \quad z_3 \quad w_1 \quad w_2 \quad w_3$

$$\frac{(0-i)(w_3-w)}{(0-w)(w_3-i)} = \frac{(z_1-i)(0-z)}{(z_1-z)(0-i)}$$

$$\frac{+i(w_3-w)}{+w(w_3-i)} = \frac{(z_1-i)(+z)}{(z_1-z)(+i)}$$

$$\frac{w_3 \left(1 - \frac{w}{w_3}\right) i}{w_3 \left(1 - \frac{i}{w_3}\right) w} = \frac{z_1 \left(1 - \frac{i}{z_1}\right)}{z_1 \left(1 - \frac{z}{z_1}\right)} \frac{z}{i}$$

$$\lim_{w_3 \rightarrow \infty} \left(\frac{1 - \frac{w}{w_3}}{1 - \frac{i}{w_3}} \right) \frac{i}{w} = \lim_{z_1 \rightarrow \infty} \left(\frac{1 - \frac{i}{z_1}}{1 - \frac{z}{z_1}} \right) \frac{z}{i}$$

$$\frac{i}{w} = \frac{z}{i} \Rightarrow i^2 = wz$$

$$-1 = wz$$

$$w = \frac{-1}{z} \checkmark$$

$p \quad z = 1, 0, -1$ onto $w = i, 1, \infty$
 $z_1 \quad z_2 \quad z_3 \quad w_1 \quad w_2 \quad w_3$

$$\frac{(i-1)(w_3-w)}{(i-w)(w_3-1)} = \frac{(1-0)(-1-z)}{(1-z)(-1-0)}$$

$$\Rightarrow \frac{(i-1)(w_3-w)}{(i-w)(w_3-1)} = \frac{-1-z}{-1+z}$$

$$\frac{w_3 \left(1 - \frac{w}{w_3}\right) (i-1)}{w_3 \left(1 - \frac{1}{w_3}\right) (i-w)} = \frac{-1-z}{-1+z}$$

$$\frac{i-1}{i-w} = \frac{-1-z}{-1+z}$$

$$-i + iz + 1 + z = -1 - iz + w + wz$$

$$iz + 1 + z = -iz + w + wz$$

$$2iz + z + 1 = w + wz$$

$$\Rightarrow w(1+z) = z(1+2i) + 1$$

$$\therefore w = \frac{(1+2i)z + 1}{1+z}$$

p Find the invariant (fixed) points of the transformation

$$1 \quad w = \frac{z-1}{z+1} \quad w = \frac{z}{1} = \frac{z-1}{z+1} \Rightarrow z^2 + z = z - 1$$

$$z^2 = -1, \quad z = \pm i$$

$$2 \quad w = \frac{6z-9}{z}, \quad w = z = \frac{6z-9}{z} \Rightarrow 6z-9 = z^2, \quad z^2 - 6z + 9 = 0$$

$$(z-3)^2 = 0 \quad z = 3$$

Linear fractional transformation or Mobius transformation.

$$w = \frac{az+b}{cz+d} \quad (ad-bc \neq 0).$$

where a, b, c, d are complex or real numbers. Differentiation gives.

$$w' = \frac{ad-bc}{(cz+d)^2}$$

This motivates our requirement $ad-bc \neq 0$. It implies conformality for all z and excludes the totally unintersecting case $w' = 0$. If $ad-bc = 0$ then the case is said to be totally univalent.

Q Define conformal mapping, given an Example?

1 Find the bilinear transformation that maps z_1, z_2, z_3 on to w_1, w_2, w_3 respectively

$$z = -1, 0, 1, \text{ on to } w = 0, i, 3i$$

$$\frac{(w_1 - w_2)(w_3 - w)}{(w_1 - w)(w_3 - w_2)} = \frac{(z_1 - z_2)(z_3 - z)}{(z_1 - z)(z_3 - z_2)}$$

$$\frac{(0-i)(3i-w)}{(0-w)(3i-i)} = \frac{(-1-0)(1-z)}{(-1-z)(1-0)} \Rightarrow \frac{+i(3i-w)}{+w(2i)} = \frac{-1(1-z)}{-1-z}$$

$$\Rightarrow \frac{3i-w}{2w} = \frac{1-z}{1+z}$$

$$\Rightarrow (3i-w)(1+z) = 2w(1+z)$$

$$3i + 3iz - w - wz = 2w + 2wz$$

$$3i + 3iz = 2w + 2wz + w + wz$$

$$3i + 3iz = 3w + 3wz \Rightarrow 3i(1+z) = 3w + 3wz$$

$$3i + 3iz = 3w(1+z)$$

$$3i(1+z) = w(3-z)$$

$$w = \frac{-3i(z+1)}{z-3}$$

P $z = 0, -i, -1$ onto $w = i, 1, 0$
 $z_1 \quad z_2 \quad z_3 \quad w_1 \quad w_2 \quad w_3$

$$\frac{(w_1 - w_2)(w_3 - w)}{(w_1 - w)(w_3 - w_2)} = \frac{(z_1 - z_2)(z_3 - z)}{(z_1 - z)(z_3 - z_2)}$$

$$\frac{(i - 1)(0 - w)}{(i - w)(0 - 1)} = \frac{(0 + i)(-1 - z)}{(0 - z)(-1 + i)}$$

$$\frac{(i - 1)(-w)}{(i - w)(-1)} = \frac{i(-1 - z)}{-z(-1 + i)}$$

$$\frac{(i - 1)w}{i - w} = \frac{i(-1 - z)}{-z(-1 + i)}$$

$$\Rightarrow -z(-1 + i)(i - 1)w = i(-1 - z)(i - w)$$

$$\Rightarrow -zw(-i + 1 + i^2 - i) = i(-i + w - zi + zw)$$

$$\Rightarrow -zw(-i + 1 - 1 - i) = i(-i + w - zi + zw)$$

$$\Rightarrow -zw(-2i) = i(-i + w - zi + zw)$$

$$\Rightarrow zw2i = i(-i + w - zi + zw)$$

$$\Rightarrow 2zw = -i + w - zi + zw$$

$$2zw = w + zw - i - zi$$

$$2zw = w + zw - i - zi$$

$$2zw - w - zw = -i - zi$$

$$zw - w = -i - zi$$

$$w(z - 1) = -i - zi$$

$$w = \frac{-i - zi}{z - 1} = -i \left(\frac{z + 1}{z - 1} \right) \checkmark$$

$$\begin{aligned} zw - w &= (z + 1)i \\ (z - 1)w &= (z + 1)i \\ \frac{(1 + z)i}{z - 1} &= w \end{aligned}$$